

# Hein Zarni Aung

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## EDUCATION

### Stony Brook University

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Software Security, Systems Fundamentals I & II, Algorithms, Full Stack Development

Expected Graduation: May 2027

## RESEARCH

### Undergraduate Research Assistant — Ethos Lab, SBU

PI: Abisheka Pitumpe | Team size: 2

Sep 2025 – Dec 2025 (paused)

- Built an **automated verification workflow** using Selenium to monitor newly issued SSL/TLS certificates in real time, scraping and analyzing HTML, CSS, JS, and URL data to flag anomalous domains impersonating EZ-Pass tolling services.
- Used Python to filter and triage flagged sites, treating **false positives and edge cases** as first-class review criteria; auto-captured homepage screenshots of promising candidates to speed up manual review.

## INTERNSHIPS

### Software Engineering Intern — Clustr

Cross-platform React Native social/scheduling app | ~8-person dev team

Jan 2026 – Sept 2026 (expected)

- Contributed to Clustr, a cross-platform React Native social/scheduling app, delivering 50+ commits across backend and frontend features.
- Built server-side **RESTful API endpoints** using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Collaborated with an 8-person, multi-developer team via Git, regularly merging and resolving conflicts across feature branches to keep modules in sync with the shared codebase.

### Software Engineering Intern — Prenora Dynamics LLC

~8-person team

Jun 2026 – Sept 2026 (expected)

- Own end-to-end backend design decisions for an in-house, in-platform video chat feature, researching and evaluating **communication SDKs and protocols** to replace reliance on third-party platforms (e.g., Zoom, Google Meet).

## PROJECTS

### TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon)

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

Apr 2026

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time/compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Collaborated in a 4-person team; **containerized the backend in Docker and deployed it on GCP** with Cloud Firestore, alongside a React/Tailwind frontend and ElevenLabs-generated voice feedback.

### Discord Voice Channel Activity Monitor

Node.js, Discord.js, Cloud

- Built a Node.js/Discord.js bot using **event-driven, publish/subscribe-style listeners** to notify users in real time when someone joins a voice channel, solving a scheduling problem for a friend group spread across different schedules.
- Managed production deployment and configuration on a cloud server using PM2, supporting customizable DM notifications and per-server/per-user settings at 50 concurrent active users.

## SKILLS

Languages: Python, C, JavaScript, TypeScript, Java

Backend: Node.js, Express.js, FastAPI

Frontend: React, React Native, Next.js

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium